| | |
|---|---|
| **Status** | Finished |
| **Started** | Thursday, 30 October 2025, 12:01 PM |
| **Completed** | Thursday, 30 October 2025, 12:23 PM |
| **Duration** | 22 mins 19 secs |
| **Marks** | 2.00/3.00 |
| **Grade** | **2.00** out of 3.00 (**66.68**%) |

Information

# Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Consider the recurrence relation

$$4 \cdot a_n - 12 \cdot a_{n-1} - 12 \cdot a_{n-2} = 0\,.$$

Let $\{a_n\}_n$ and $\{b_n\}_n$ be two numerical sequences. We say that $a_n$ *is a solution* if the sequence $\{a_n\}_n$ satisfies the recurrence relation, and likewise for $b_n$.
Select every statement below which is true, and leave the false statements unchecked.

- [x] If $a_n$ and $b_n$ are solutions, and if $a_1$=$b_1$ and $a_3$=$b_3$, then $a_n$=$b_n$ for each n≥1.
- [ ] The zero sequence $a_n$=0 is not a solution.
- [x] If $a_n$ is a solution, and $a_1$=$a_2$=0, then $a_n$=0 for each n≥1.
- [x] If $a_n$ is a solution, then $b_n$=4$a_n$ is also a solution.
- [ ] If $a_n$ is a solution, and $a_0$=-1 and $a_1$=-2, then there exists at least one index n such that $a_n$ is not an integer.
- [ ] If $a_0$=0, $a_1$=3, then $a_n$ can not be a solution.
- [ ] None of the above

Question **2**

Incorrect

Mark 0.00 out of 1.00

Consider G = ⟨ x ⟩ ⊆ $S_7$, the subgroup of the symmetric group of permutations of 7 symbols, generated by a permutation **x** of order 3.

Recall that a subgroup H ⊆ G is called *non-trivial* if H ≠ {e}, and *proper* if H ≠ G.

Select every statement below which is true, and leave the false statements unchecked.

- [ ] The group G is infinite.
- [x] The group G is cyclic.
- [ ] The group G has a non-trivial and proper subgroup.
- [ ] The order of G is 6.
- [x] There exist elements g,h ∈ G such that gh ≠ hg.
- [ ] None of the above

Question **3**

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing products and inverses in the group $(P_{48}, \bullet_{48})$ of integers and multiplication modulo $48$.

$37 \bullet_{48} 41$ = 29

Your last answer was interpreted as follows:

$$29$$

$41 \bullet_{48} 29$ = 37

Your last answer was interpreted as follows:

$$37$$

$29 \bullet_{48} 35$ = 7

Your last answer was interpreted as follows:

$$7$$

$37^{-1}$ = 13

Your last answer was interpreted as follows:

$$13$$

Question **4**

Correct

Mark 0.00 out of 0.00

# Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: $537$

Signing code: 27592

Your last answer was interpreted as follows:

$$27592$$

◄ Test 2 (topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations)

Jump to...